

Questions

Forwards/Futures

Question

3. Suppose that in 3 months the cost of a pound of Colombian coffee will be either \$1.25 or \$2.25. The current price is \$1.75 per pound.
- (a) What are the risks faced by a hotel chain who is a large purchaser of coffee?
 - (b) What are the risks faced by a Colombian coffee farmer?
 - (c) If the delivery price of coffee turns out to be \$2.25, should the farmer have forgone entering into a futures contract? Why or why not?

3. (a) On one hand, the hotel chain faces the risk that the price at which it purchases coffee will rise. On the other hand, the hotel chain can pass along these costs to the consumers of the coffee, so it is not clear to what extent the hotel chain will be affected by an increase in coffee prices.

The hotel chain's ability to pass along rising costs to its customers is a type of a *natural hedge*. A company's failure to recognize its natural hedges can lead to "hedges" that actually increase risk (e.g., a hotel chain that locks in its costs by buying forward coffee might be at a competitive disadvantage if coffee prices fall).

- (b) The coffee farmer has no natural hedge and is at risk if the price of coffee falls. The coffee farmer may wish to sell forward coffee.
- (c) When entering the contract, the coffee farmer presumably understood the risk that a hedge may lock in a lower selling price if coffee prices rise above the futures price. The fact that the farmer lost the upside while protecting the downside does not indicate that there were "losses" or that the farmer was wrong.

Question

4. Consider a 6-month forward contract (delivers one unit of the security) on a security that is expected to pay a \$1 dividend in three months. The annual risk-free rate of interest is 5%. The security price is \$20. What forward price should the contract stipulate, so that the current value of entering into the contract is zero?

Solution

4. (a) Assume the Yield Curve is flat at 5%

$$\text{Price of forward} = \$20 * (1 + 5\%)^{.5} - \$1 * (1 + 5\%)^{.25} = \$19.482$$

The second terms takes into account that you don't receive the \$1 dividend if you enter into the futures today.

Options

Question

1.5 Options

1. A stock price is currently \$50. It is known that at the end of two months it will be either \$53 or \$48. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a two-month European call option with a strike price of \$49?

Solution

1. At the end of two months the value of the option will be either $C_u = \$4$ (if the stock price is \$53) or $C_d = \$0$ (if the stock price \$48). Moreover, $u = \frac{53}{50} = 1.06$, $d = \frac{48}{50} = .96$ and $R = e^{10\% \times \frac{2}{12}} = 1.017$ so that the risk-neutral probability of “up” is $q = \frac{R-d}{u-d} = 0.57$. The value of the option is therefore

$$C = \frac{1}{R} \times [qC_u + (1 - q)C_d] = \$2.235.$$

Question

2. A stock price is currently \$80. It is known that at the end of four months it will be either \$75 or \$85. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a four-month European put option with a strike price of \$80?

Answer

2. At the end of four months the value of the option will be either $P_d = \$5$ (if the stock price is \$75) or \$0 (if the stock price $P_u = \$85$). Moreover, $u = \frac{85}{80} = 1.063$, $d = \frac{75}{80} = .94$ and $R = e^{5\% \times \frac{4}{12}} = 1.017$ so that the risk-neutral probability of “up” is $q = \frac{R-d}{u-d} = 0.634$. The value of the option is therefore

$$P = \frac{1}{R} \times [qP_u + (1 - q)P_d] = \$1.797$$

Question

3. Today's price of three traded call options on BackBay.com, all expiring in one month, are as follows:

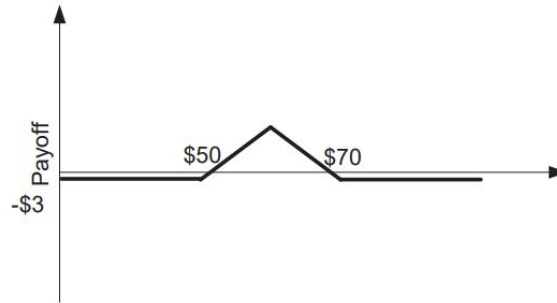
Strike Price	Option Price
\$50	$\$7\frac{1}{2}$
60	\$3
70	$\$1\frac{1}{2}$

You are considering buying a “butterfly spread” consisting of the following positions:

- Buy 1 call at strike price of \$50
 - Sell (write) 2 calls at strike price of \$60
 - Buy 1 call at strike price of \$70.
- (a) Plot the payoff of your total position for different values of the stock price on the maturity date.
- (b) What is the dollar investment required to establish the spread?
- (c) For what stock prices on the maturity date will you be making an overall profit?

Solution

3. (a) The payoff is as follows:



- (b)

$$\begin{aligned}\text{Cost} &= (\text{Long 50 Call}) - 2 \times (\text{Short 60 Call}) + (\text{Long 70 Call}) \\ \$3.00 &= \$7.50 - 2 \times \$3.00 + \$1.50\end{aligned}$$

This position requires an initial investment of \$3.00

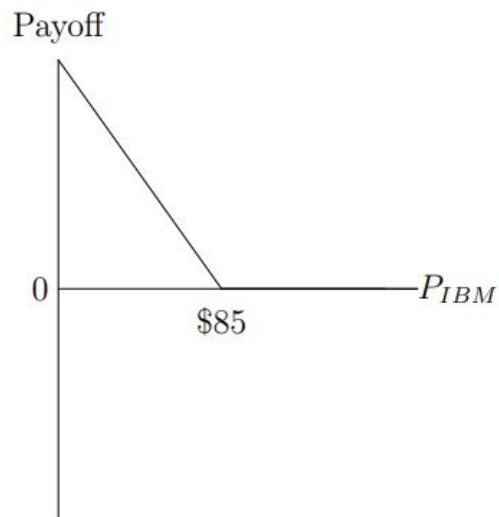
- (c) There will be a positive profit for stock prices at option maturity between \$53.00 and \$67.00

Question

7. IBM shares are now traded at \$80.00. The term structure of interest rates is flat at 5%.
- (a) Plot the terminal payoff from a European put option on 1 share of IBM with a exercise price of \$85 and a maturity of 3 months (not including the price of the option).
 - (b) Suppose that you purchase the put option at \$4.00 from the market. Specify the ranges of IBM share price at the options maturity date for which you will be making a net profit.

Solution

7. (a) Payoff diagram:



(b) You will make money as long as $P_{IBM} \leq \$85 - FV(\$4) = \$85 - \$4.05 = \$80.95$

Risk/Portfolio choice

Question

1. True or false or "it depends"?

- (a) Briefly explain or qualify your answer: diversification can reduce risk only when asset returns are negatively correlated.
- (b) If the returns on all risky assets in the world were uncorrelated with each other, the expected return of each risky asset should be the same.

Answer

1. (a) False: As long as asset returns are not perfectly positively correlated, diversification will reduce risk.
- (b) True: If returns are completely uncorrelated, then all risk is non-systematic risk that can be diversified away. So the expected return for all assets should be the same - the risk-free rate of return.

Question

2. True or false or "it depends"? Optimal portfolios should exclude individual assets whose expected return and risk (measured by its standard deviation) are dominated by other available assets.

Answer

2. False. Standard deviation includes both systematic and idiosyncratic risks. An asset that appears dominated may be correlated with other assets in such a way that makes it useful (i.e., variance reducing) in a diversified portfolio.

3. Is the following statement true or false? Explain. As more securities are added to a portfolio, total risk would typically be expected to fall at a decreasing rate.

Answer

3. True. Only a few securities are needed to reduce a significant portion of idiosyncratic risks. Benefits to diversification become smaller afterwards. Suppose we have an equally-weighted portfolio of n securities that have identical expected return and variance (σ^2), but are uncorrelated with each other. The portfolio variance is then $\frac{\sigma^2}{n}$, which is a decreasing and convex function of n .

Question

4. You need to invest \$10M in two assets: a risk-free asset with an expected return of 5% and a risky asset with an expected return of 12% and a standard deviation of 40%. You face a cap of 30% on the portfolio's standard deviation (the “risk budget”). What is the maximum expected return you can achieve on your portfolio?

Answer

4. 30% portfolio standard deviation $\Rightarrow \frac{30\%}{40\%} = 0.75$ in risky and 0.25 in riskfree $\Rightarrow$ portfolio expected return is $(0.25)(5\%) + (0.75)(12\%) = 10.25\%$.

Question

5. Are the following statements true, false or uncertain? Briefly explain your answer.
- (a) Diversification over a large number of assets completely eliminates risk.
 - (b) Diversification works only when asset returns are uncorrelated.
 - (c) A stock with high standard deviation may contribute less to portfolio risk than a stock with lower standard deviation.
 - (d) Diversification reduces the expected return on the portfolio as its risk decreases.

Answer

5. (a) False. Diversification cannot eliminate systematic risk.
- (b) False. Diversification works whenever asset returns are not perfectly correlated.
- (c) True. The contribution of a stock to portfolio risk depends on its systematic risk, not total risk.
- (d) False. Diversification may reduce portfolio risk while keeping expected return constant (e.g., a portfolio of many securities with identical return/risk but imperfect correlation).

Question

9. You have decided to invest all your wealth in two mutual funds: A and B. Their returns and risks are as follows:

- the mean returns are $\tilde{r}_A = 15\%$ and $\tilde{r}_B = 11\%$
- the covariance matrix is

	r_A	r_B
r_A	.04	.025
r_B	.025	.032

You want your total portfolio to yield a return of 12%. What proportions of your wealth should you invest in A and B? What is the standard deviation of the return on your portfolio?

Answer

9. Let x be the proportion of B . The expected portfolio return is $11x + 15(1 - x) = 12$, which gives $x = \frac{3}{4} = 0.75$. So you should invest 75% of your wealth in B and 25% in A . The variance is then

$$\begin{aligned}\text{var} [(1 - x)A + xB] &= (1 - x)^2\sigma_A^2 + x^2\sigma_B^2 + 2x(1 - x)\sigma_{AB}. \\ \sigma^2 &= 0.25^2 \times 0.04 + 0.75^2 \times 0.032 + 2 \times 0.25 \times 0.75 \times 0.025 = 0.030.\end{aligned}$$

CAPM

Question

1. What is the beta of a portfolio with $E(r_p) = 18\%$, if $r_f = 6\%$ and $E(r_M) = 14\%$?

Answer

$$E(r_p) = r_f + \beta [E(r_m) - r_f]$$

$$18 = 6 + \beta (14 - 6)$$

$$\beta = 12/8 = 1.5$$

Question

2. You are a consultant to a large manufacturing corporation that is considering a project with the following net after-tax cash flows (in millions of dollars):

Years from Now	After-Tax Cash Flow
0	-40
1 - 10	15

The project's beta is 1.8. Assuming that $r_f = 8\%$ and $E(r_M) = 16\%$, what is the net present value of the project? What is the highest possible beta estimate for the project before its NPV becomes negative?

Answer

2. The appropriate discount rate for the project is:

$$r_f + \beta [E(r_m) - r_f] = 8 + 1.8(16 - 8) = 22.4\%.$$

Using this discount rate,

$$\begin{aligned} NPV &= -40 + 15 \left[\frac{1}{.224} - \frac{1}{(.224)(1.224^{10})} \right] \\ &= 18.09. \end{aligned}$$

Using Excel, we find that the breakeven discount rate is .353. The highest value that beta can take therefore is determined by

$$\begin{aligned} 35.73 &= 8 + \beta(16 - 8) \\ \beta &= 27.73/8 = 3.467. \end{aligned}$$

Question

3. Are the following true or false?

- (a) Stocks with a beta of zero offer an expected rate of return of zero.
- (b) The CAPM implies that investors require a higher return to hold highly volatile securities.
- (c) You can construct a portfolio with a beta of 0.75 by investing 0.75 of the investment budget in bills and the remainder in the market portfolio.

Answer

3. (a) False. $\beta = 0$ implies $E(r) = r_f$, not zero.
- (b) False. Investors require a risk premium only for bearing systematic risk (undiversifiable or market) risk.
- (c) False. 75% of your portfolio should be in the market, and 25% in bills. Then,

$$\beta_p = .75 \times 1 + .25 \times 0 = .75.$$

Question

4. Assume that the risk-free rate of interest is 6% and the expected rate of return on the market is 16%. A share of stock sells for \$50 today. It will pay a dividend of \$6 per share at the end of the year. Its beta is 1.2. What do investors expect the stock to sell for at the end of the year?

Answer

4. Since the stock's beta is equal to 1.2, its expected rate of return should be equal to $6 + 1.2(16 - 6)$ or 18%.

$$E(r) = \frac{D + P_1 - P_0}{P_0}$$

$$.18 = \frac{6 + P_1 - 50}{50}$$

$$P_1 = \$53.$$

Capital Budgeting

Question

1. Table A gives investments, NPVs, IRRs and the first three years' cash flow for several capital investment projects. Each project's cash flows continue for several more years, longer for some projects than others. The cost of capital is 12% for all projects.

Table A (figures in millions).

Project	Invest in 2000	C_1	C_2	C_3	NPV	IRR
A	100	20	20	20	57	17.8
B	200	0	20	40	64	14.5
C	50	20	20	20	41	37.8
D	75	-10	10	30	0	12
E	30	-10	5	7	-3	11
F	10	3	4	5	5.5	30.2

Projects A and B are mutually exclusive - your firm can take only one. The projects are discrete - you cannot make partial investments in any project.

- (a) Suppose the firm rejects all projects with payback periods greater than 3 years. What is the NPV from following this policy?
- (b) A manager defends the decisions in part a as a way to avoid taking on risky projects. Does this defense make sense?
- (c) Which project would you choose, A or B?
- (d) Suppose that the firm now identifies a new project AA with exactly the same cash flows, NPV and IRR as project A. Does the opportunity to invest in AA change your answer to part c?
- (e) Suppose the firm has only \$200 million to invest - a fixed capital constraint. Which projects would you undertake? (Ignore project AA.)
- (f) Now the firm negotiates a line of credit that allows it to borrow up to \$100 million at 8%. Would access to additional debt capital at a cost of 8% change your answers to questions c, d or e?

Explain each answer briefly.

Answer

1. (a) Projects A,B, D, E all have payback period of more than 3 years. So you will only take C & F for a total NPV of 46.5
- (b) No. Two possible answers:
 - It is not the manager's role to avoid taking risk. They should seek to create maximum value from their projects, that is maximize NPV. Investor will adjust their portfolio to reflect their risk tolerance.
 - All projects here have a cost-of-capital of 12%. If this is calculated correctly, all projects are equally risky in terms of systematic risk. The rest is diversifiable risk and investor can easily get rid of those by holding a well diversified portfolio.
- (c) B, for higher NPV.
- (d) No. I will still prefer B over A. Of course, I prefer the combination of A and AA over B, but that's not the question here.
- (e) You basically have 2 choices: Take B, which use up all your capital, or invest in A+C+F. The latter give you higher NPV at $57+41+5.5 = 103.5$.
- (f) My answers to c and d stay the same. The borrowing cost does not reflect the true cost of capital. However, as long as the debt is issued at fair value, I can utilize it and expand my investment choice to 300m. Then I can take both projects A & B for a NPV of 121.

Question

4. The DEF Corporation is trying to decide whether to undertake an expansion of its production facilities. The expansion will cost \$8.5 million, to be paid immediately. After tax cash flows generated by the expansion are projected to be \$1 million next year, and will be growing indefinitely with inflation at 2.5% per year. Assume that the CAPM holds, the beta of DEF assets is 1.2, the riskless rate is 5% per year (and the yield curve is flat at this rate) and that the expected return on the market portfolio is 12%. Should DEF undertake the expansion?

Answer

4. The appropriate discount rate is $5\% + 1.2(12\% - 5\%) = 13.4\%$.

$$\text{NPV} = -8.5 + \frac{1}{0.134 - 0.025} = \$0.67 \text{ million.}$$

DEF should undertake the expansion.

